

● HARD

● GEOMETRY AND TRIGONOMETRY

Area and volume

03

9 questions · ~14 min

Q1

GRID-IN GEOMETRY AND TRIGONOMETRY

A cube has a volume of 474,552 cubic units. What is the surface area, in square units, of the cube?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2

GEOMETRY AND TRIGONOMETRY

A hemisphere is half of a sphere. If a hemisphere has a radius of 27 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

- A. 1,500
- B. 6,100
- C. 30,900
- D. 41,200

Q3

GEOMETRY AND TRIGONOMETRY

A right circular cone has a volume of $\frac{1}{3}\pi$ cubic feet and a height of 9 feet. What is the radius, in feet, of the base of the cone?

- A. $\frac{1}{3}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\sqrt{3}$
- D. 3

The circumference of the base of a right circular cylinder is 20π meters, and the height of the cylinder is 6 meters. What is the volume, in cubic meters, of the cylinder?

- A. 60π
- B. 120π
- C. 600π
- D. $2,400\pi$

Q5

GRID-IN

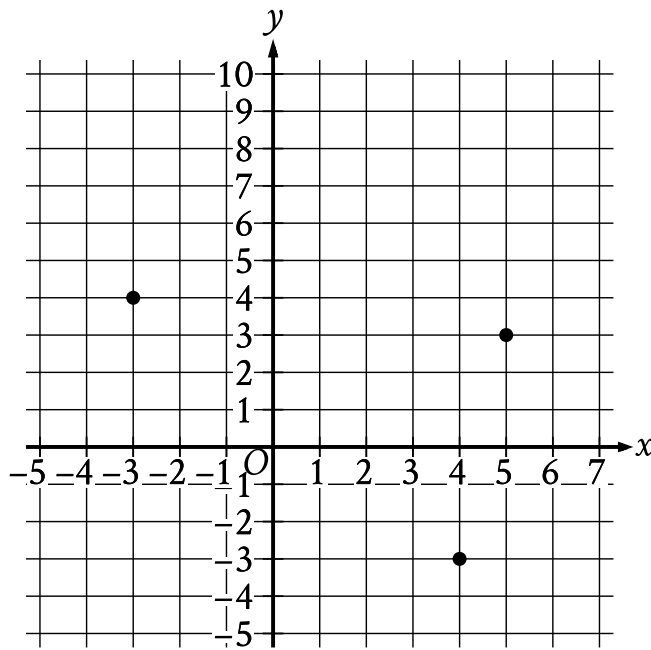
GEOMETRY AND TRIGONOMETRY

Square A has side lengths that are 166 times the side lengths of square B. The area of square A is k times the area of square B. What is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



- Moving from left to right, the points have the following coordinates:
 - (negative 3 comma 4)
 - (4 comma negative 3)
 - (5 comma 3)

What is the area, in square units, of the triangle formed by connecting the three points shown?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5

6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

GEOMETRY AND TRIGONOMETRY

Two identical rectangular prisms each have a height of 90 centimeters cm. The base of each prism is a square, and the surface area of each prism is K cm². If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47}K$ cm². What is the side length, in cm, of each square base?

- A. 4
- B. 8
- C. 9
- D. 16

Q8

GEOMETRY AND TRIGONOMETRY

Circle A has a radius of $3n$ and circle B has a radius of $129n$, where n is a positive constant. The area of circle B is how many times the area of circle A ?

- A. 43
- B. 86
- C. 129
- D. 1,849

Q9

GEOMETRY AND TRIGONOMETRY

The dimensions of a right rectangular prism are 4 inches by 5 inches by 6 inches. What is the surface area, in square inches, of the prism?

- A. 30
- B. 74
- C. 120
- D. 148